SUPPORTING ADULT GUIDANCE

Solving by elimination

Equations, Inequalities and Simultaneous Relationships · Lesson 8. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson develops the **elimination** method: add or subtract the equations (scaling a row first when needed) so one variable cancels, then solve and back-substitute. Following Gerry's ordering, the easier **add** case (opposite signs) leads, then **subtract** (same signs), then **scale-then-eliminate**. It is the second exact method for simultaneous equations, complementing substitution.

Driving Question: *How do I combine two equations to remove an unknown?*

The mathematics, in plain terms

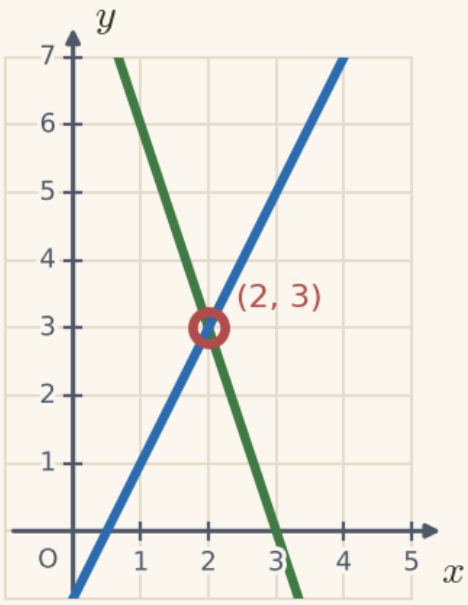
Adding or subtracting two equations yields another valid equation. Choosing the operation so a variable's terms sum to zero (opposite signs → add; equal signs → subtract) removes that variable. When coefficients differ, multiplying an equation through by a constant (which preserves its truth) makes them match. The reduced single-variable equation is solved and back-substituted; both originals are checked.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Opposite signs — add

Solve $3x + y = 9$ and $2x - y = 1$. The y-terms are $+y$ and $-y$ — opposite.

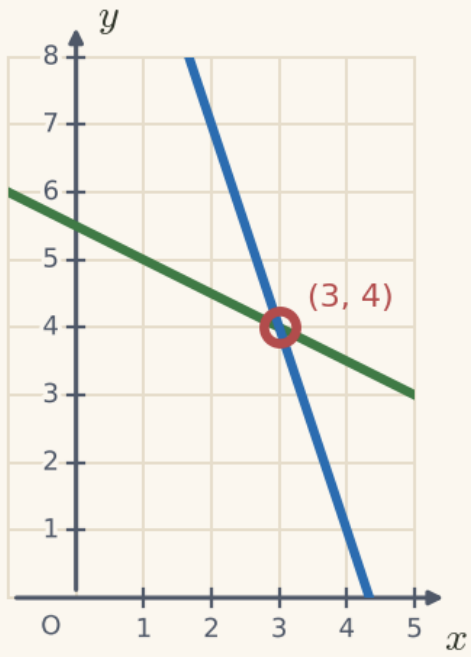


1. Add the two equations — the y cancels.	$(3x + y) + (2x - y) = 9 + 1$
2. One unknown remains.	$5x = 10$, so $x = 2$
3. Back-substitute to find y.	$3\times2 + y = 9$, so $y = 3$
4. Check in both.	$3\times2 + 3 = 9$ ✓ and $2\times2 - 3 = 1$ ✓

(2, 3) — opposite signs mean adding cancels y; then $x = 2$ and $y = 3$, true in both.

Example 2 — No match — scale, then subtract

Solve $x + 2y = 11$ and $3x + y = 13$. Nothing matches yet.



1. Multiply the second equation by 2 so the y-terms match.	$6x + 2y = 26$
2. Now both have $+2y$ — subtract to cancel y.	$(6x + 2y) - (x + 2y) = 26 - 11$
3. Solve the result.	$5x = 15$, so $x = 3$
4. Back-substitute and check.	$x + 2y = 11 \rightarrow y = 4$; both true ✓

(3, 4) — scaling the second equation by 2 lets subtraction cancel y; $x = 3$, $y = 4$.

Add (opposite) first, then subtract (same), then scale

Teach the decision from the signs, easiest first: opposite signs invite **adding** — the cancellation is very visible — so lead with that; then same signs and **subtracting**; then the case where you must **scale a row** before either works. Common slips: subtracting term by term with sign errors (bracket the second equation), and forgetting to back-substitute for the second variable.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Add two opposite-sign equations and watch a variable vanish.	~8 min

Movement	Work the full method: add / subtract / scale, then solve and check.	~13 min
Reflection	Decide from the signs alone: add or subtract to cancel?	~9 min
Creativity	Scale a row so coefficients match, then eliminate.	~8 min
Rest	A short pause on combining two equations head-on.	~2 min
Nutrition	Intellectual: leverage — acting on two equations at once.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding when the signs are the same (so nothing cancels).
Check the matching variable's signs: 'Both $+y$? Then adding gives $2y$ — it doesn't cancel. Same sign means subtract.'

Sign errors when subtracting, e.g. forgetting to negate every term.
Bracket the second equation and subtract as a whole: $(3x + y) - (x + y)$. 'What happens to each term when the minus goes in?'

Trying to eliminate before coefficients match.
'Do any coefficients match yet? If not, multiply one equation so a variable lines up first.'

Solving for one variable and stopping.
A solution is a pair. 'You have x — put it back into an original equation to find y, then check both.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: adding and subtracting like terms, scaling an equation, and back-substituting to find the second variable. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner adds two same-sign equations, or drops a sign when subtracting. The aim is a question that returns the decision to them.

- "Look at the y-terms — same sign or opposite? So do we add or subtract?"
- "Bracket the second equation: $(3x + y) - (x + y)$. What is each term after the minus?"

- “Do any coefficients match yet? If not, what could you multiply to make them match?”
- “You found x — how do you get y, and does the pair work in both equations?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Substitution or elimination — which should they use?

Either works. Elimination is often quickest when both equations are in the form ax + by = c with a variable that already matches or easily scales; substitution suits a variable already made the subject. Let learners choose and justify.

Why lead with adding?

Because a $+y$ and a $-y$ cancelling on adding is the most visible case — learners spot it readily, which builds confidence before the subtract and scale cases.

Do the answers have to be whole numbers?

No — the stretch gives $x = 0.5$. Elimination handles fractional solutions exactly.

What if I have to scale both equations?

Sometimes you multiply each equation by a different number so a variable matches (e.g. ×3 and ×2). The principle is the same — make one variable's coefficients equal, then add or subtract.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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